

Christopher Nellis

Work Experience

Duke Energy

Principal Software Engineer

Sep 2025 — Present

- Founding member of Software Engineering Council, a cross-functional group focused on developing, reviewing, and enforcing software engineering standards across the enterprise
- Served as tech lead for the implementation of Autho for customer authentication across all Duke Energy customer-facing applications, collaborating with CyberSecurity, legal, and various platform teams
- Designed, developed, and implemented Autho Actions to enhance and secure customer authentication flows according to requirements
- Served as subject matter expert for additional Autho tenants across the enterprise
- Developed Aurora Postgres database standards across the the enterprise including credential rotation, audit logging, and upgrade procedures
- Implemented SSO for Apple's AppStoreConnect platform across Duke Energy
- Consult with CyberSecurity teams to disposition CodeQL findings, setting remediation plans or flagging false positives as necessary
- Partner with Akamai firewall team to configure integrations for Autho customer mobile app teams during incidents, evaluate proposed firewall changes, and analyze network traffic during incidents

Lead Software Engineer

Jun 2021 — Sep 2025

- Pioneered usage of ECS and Fargate at the company, becoming the first team to use those technologies
- Established patterns and best practices for deploying to ECR
- Migrated SQL Server Database to Aurora Postgres within AWS
- Architected system to send push notifications within AWS, integrating with Firebase Cloud Messaging
- Regularly communicate with product owners, non-technical stakeholders, peers, and developers to ensure a shared vision and alignment

Senior Software Engineer

Sep 2019 — Jun 2021

- Led backend development throughout SAP Billing System consolidation
- Mentored new backend developers on the team
- Advocated for test-driven development and unit testing across the backend platform
- Navigated the migration of runtime environments from Pivotal Cloud Foundry to Nirmata

Software Engineer

Oct 2017 — Sep 2019

- Developed initial feature set of the customer mobile app
- Coordinated deployment across multiple teams for initial customer mobile app release
- Instituted coding challenges for hiring processes
- Built customer recommendation engine API layer with Spring

Revature

Software Developer

Aug 2016 — Oct 2017

- Participated on development teams for internal projects
- Presented technical solutions to colleges and trainers

UNC Office of Institutional Research and Assessment

Student Worker

May 2015 — Aug 2016

- Built tool to upload accreditation evidence and track approval status
- Managed HTML copy of accreditation report
- Created tool to translate addresses into latitude and longitude coordinates using Google Maps API
- Wrote PowerShell script to rename multiple files at once

Skills

- **Spring Boot Development:** Springcloud for AWS Integrations, Spring Web, Spring Security, Spring Data, Springdoc OpenAPI, Hibernate, Lombok
- **AWS Services:** IAM, Elastic Container Service, Fargate, Elastic Container Registry, API Gateway, Lambda, Cloudwatch, ElastiCache, Aurora, DynamoDB, Secrets Manager, Route53, Simple Queue Service, Simple Storage Service, Elastic Load Balancer, Database Migration Service
- **Enterprise Tools:** Terraform, Akamai, JIRA, GitHub, GitHub Actions for CI/CD, CodeQL, Dependabot, Flyway, Dynatrace, Hashicorp Vault

Education

University of North Carolina - Chapel Hill

BS in Information Science

May 2016

- Programming in Python
- Relational Databases
- Web Development
- Systems Analysis and Design
- Text Mining